Load Dataset

```
url = "https://raw.githubusercontent.com/YBI-Foundation/Dataset/main/Fish.csv"

df = pd.read_csv(url)

df.head()
```

	Category	Species	Weight	Height	Width	Length1	Length2	Length3
0	1	Bream	242.0	11.5200	4.0200	23.2	25.4	30.0
1	1	Bream	290.0	12.4800	4.3056	24.0	26.3	31.2
2	1	Bream	340.0	12.3778	4.6961	23.9	26.5	31.1
3	1	Bream	363.0	12.7300	4.4555	26.3	29.0	33.5
4	1	Bream	430.0	12.4440	5.1340	26.5	29.0	34.0

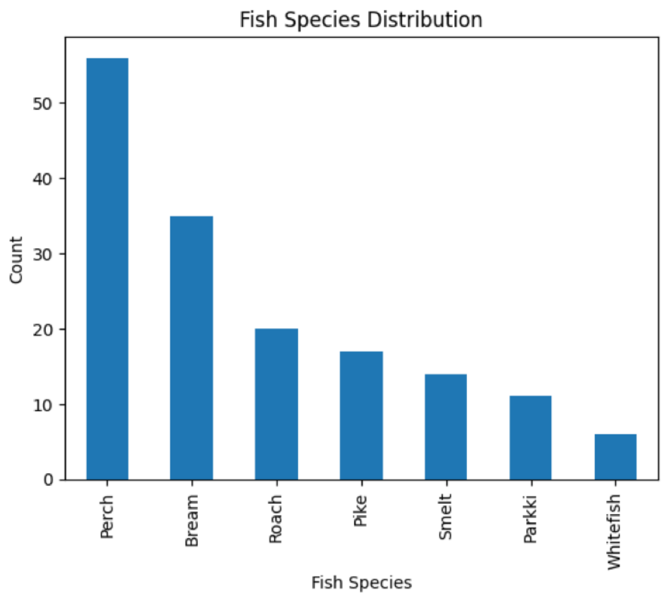
species ount graph

[16]
✓ 0s

```
df['Species'].value_counts().plot(kind='bar')

plt.xlabel("Fish Species")
plt.ylabel("Count")
plt.title("Fish Species Distribution")

plt.show()
```



Display first five rows

```
print(df.head())
```

```
   Category Species  Weight   Height   Width  Length1  Length2  Length3
0         1   Bream   242.0  11.5200  4.0200     23.2     25.4     30.0
1         1   Bream   290.0  12.4800  4.3056     24.0     26.3     31.2
2         1   Bream   340.0  12.3778  4.6961     23.9     26.5     31.1
3         1   Bream   363.0  12.7300  4.4555     26.3     29.0     33.5
4         1   Bream   430.0  12.4440  5.1340     26.5     29.0     34.0
```

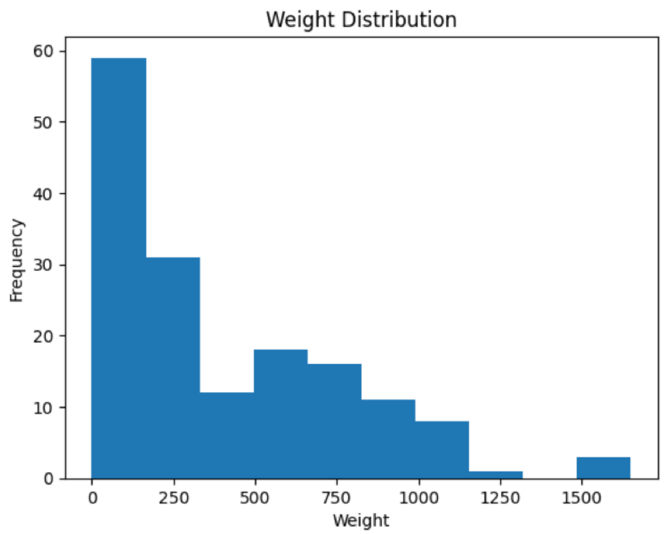
weight distribution

[17]

```
plt.hist(df['Weight'])

plt.xlabel("Weight")
plt.ylabel("Frequency")
plt.title("Weight Distribution")

plt.show()
```



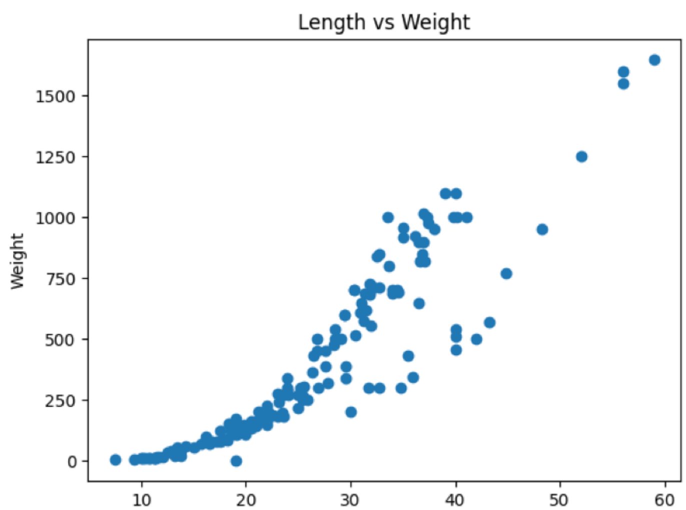
Length vs weight scatter

8]
0s

```
plt.scatter(df['Length1'], df['Weight'])

plt.xlabel("Length")
plt.ylabel("Weight")
plt.title("Length vs Weight")

plt.show()
```



Train Model

```
model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

model.fit(X_train, y_train)
```

```
RandomForestClassifier
RandomForestClassifier(random_state=42)
```

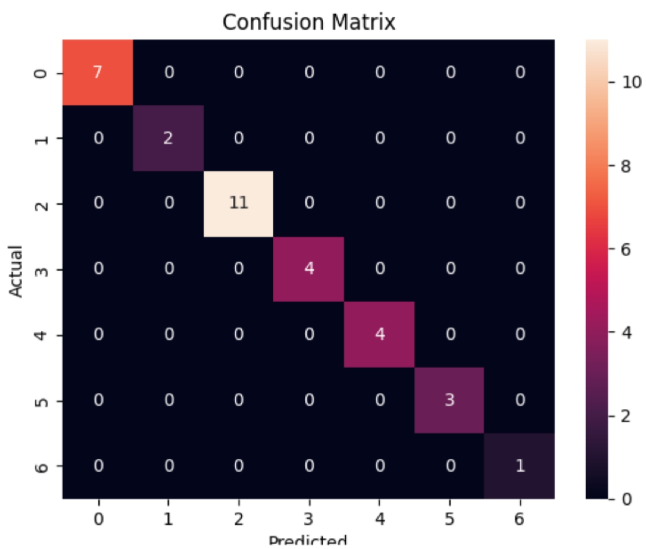
```python
from sklearn.metrics import confusion_matrix
import matplotlib.pyplot as plt
import seaborn as sns

cm = confusion_matrix(y_test,y_pred)

sns.heatmap(
    cm,
    annot=True,
    fmt='d'
)

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")

plt.show()
```



```
Accuracy: 1.0

Classification Report
              precision    recall  f1-score   support

       Bream       1.00      1.00      1.00         7
      Parkki       1.00      1.00      1.00         2
       Perch       1.00      1.00      1.00        11
        Pike       1.00      1.00      1.00         4
       Roach       1.00      1.00      1.00         4
       Smelt       1.00      1.00      1.00         3
   Whitefish       1.00      1.00      1.00         1

    accuracy                           1.00        32
   macro avg       1.00      1.00      1.00        32
weighted avg       1.00      1.00      1.00        32


Confusion Matrix
[[ 7  0  0  0  0  0  0]
 [ 0  2  0  0  0  0  0]
 [ 0  0 11  0  0  0  0]
 [ 0  0  0  4  0  0  0]
 [ 0  0  0  0  4  0  0]
 [ 0  0  0  0  0  3  0]
 [ 0  0  0  0  0  0  1]]
```

```python
plt.bar(
    X.columns,
    importance
)

plt.xlabel("Features")
plt.ylabel("Importance")
plt.title("Feature Importance")

plt.xticks(rotation=45)

plt.show()
```